Financial Stress Detection Using Reddit Data

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Overview

GOAL

Detect Financial strain in Reddit posts

Deep-learning NLP classification

DATASET

Dreaddit

Full dataset (3,553 posts) + finance subset (717 posts)

MODELS

CNN + BiLSTM

Custom Word2Vec embeddings · CNN kernels 2/3/4 · 2-layer BiLSTM

EVALUATION

Metrics + diagnostics

Accuracy / F1 / Precision / Recall · confusion matrices · training curves · error analysis

Introduction & Motivation

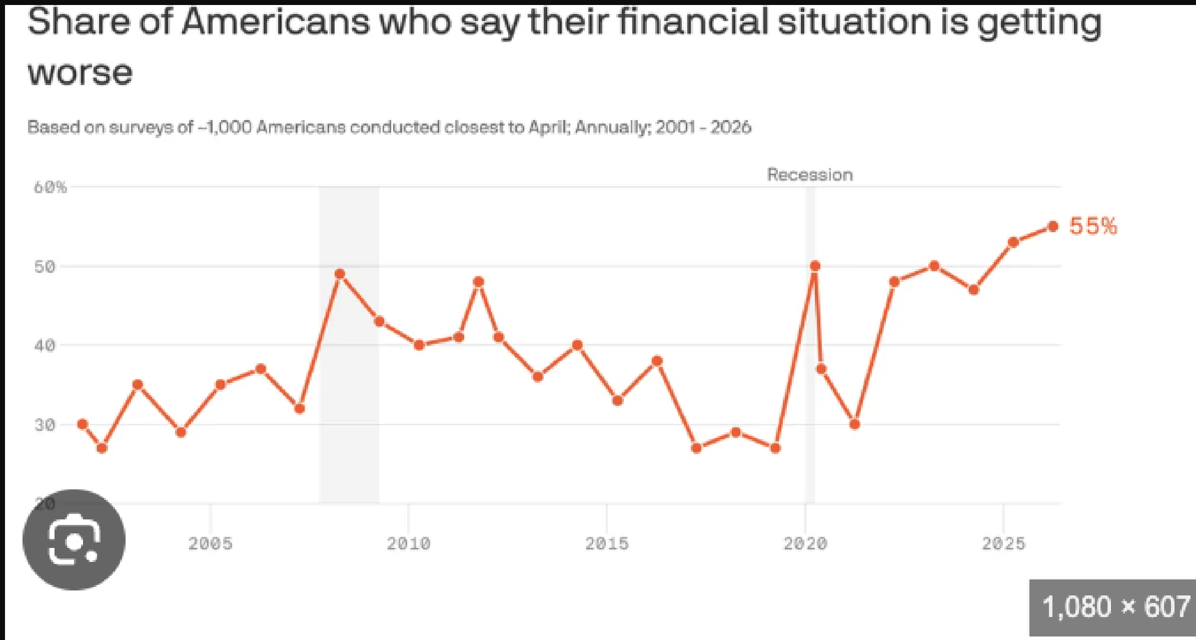
- Mental-health detection from text is a high-impact NLP problem
- Financial stress is understudied vs. anxiety / trauma research
- Reddit offers authentic, unfiltered expressions of distress
- Binary task: stressed (1) vs. not stressed (0)

RESEARCH QUESTION

Can deep-learning models reliably detect financial stress signals from social media text?

Background & Prior Research

- **Stress detection on text:** prior work on Twitter, clinical notes, online forums
- **Dreaddit (Turcan & McKeown, 2019):** first Reddit-based stress dataset
- **CNN (Kim, 2014):** strong at local n-gram patterns
- **BiLSTM:** captures long-range, bidirectional context
- **Gap:** financial stress is rarely isolated → motivates this work



Dataset & Data Preprocessing

- Built on the Dreaddit corpus, with a finance-focused subset and custom Word2Vec embeddings.

DATASET

Dreaddit

3,553 Reddit posts · 10 subreddits · finance subset of 717 posts (assistance, homeless, almosthomeless, food_pantry)

PREPROCESSING

Cleaned text

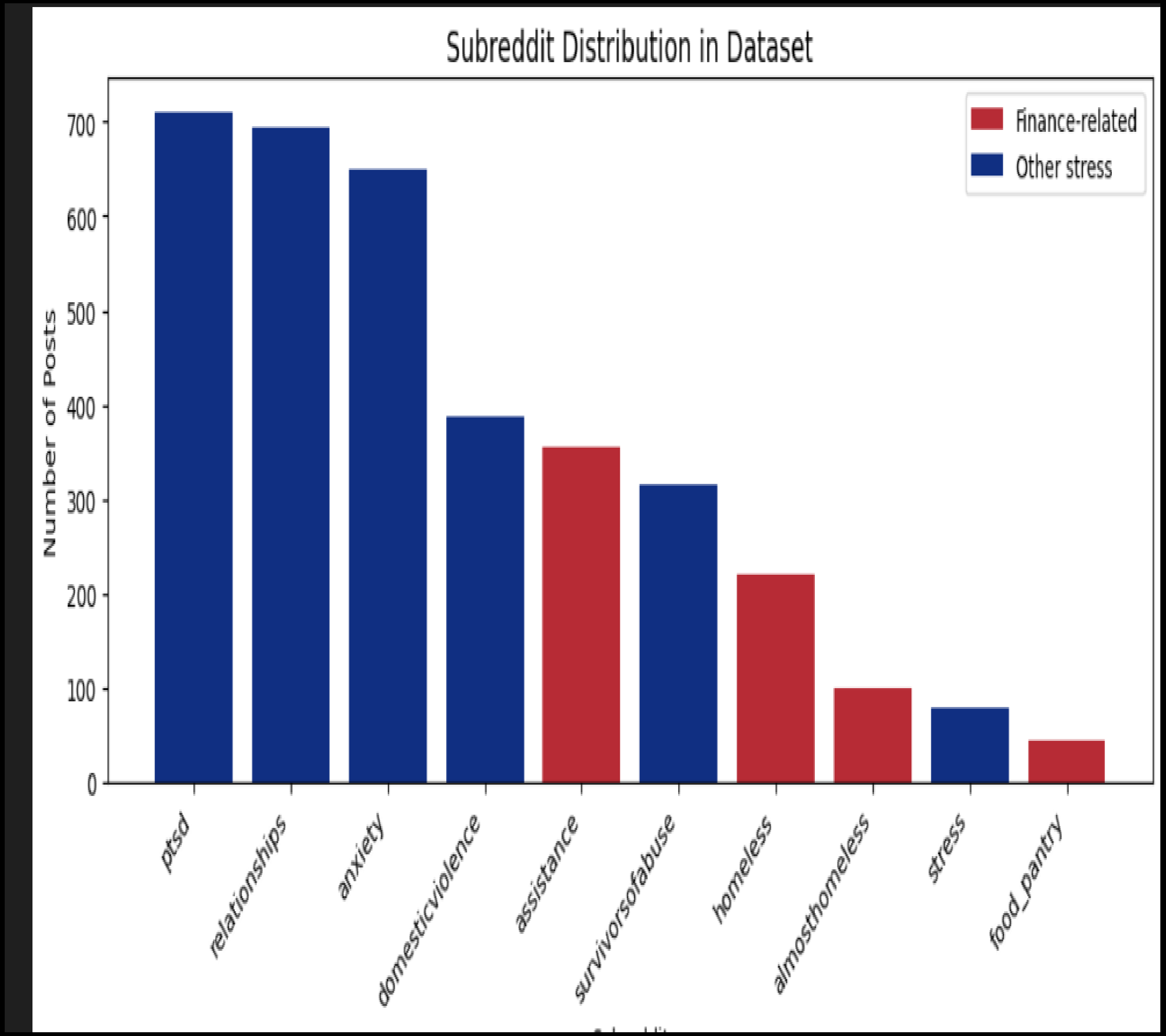
Removed URLs, punctuation, stopwords · avg 40 tokens per post

EMBEDDINGS

Custom Word2Vec

Trained on corpus · vocab 6,873 · 100 dimensions

Dataset Exploration: Where the Data Comes From



STRESS COMMUNITIES

- 10 subreddits total
- r/ptsd and r/relationships are largest (700+ posts each)

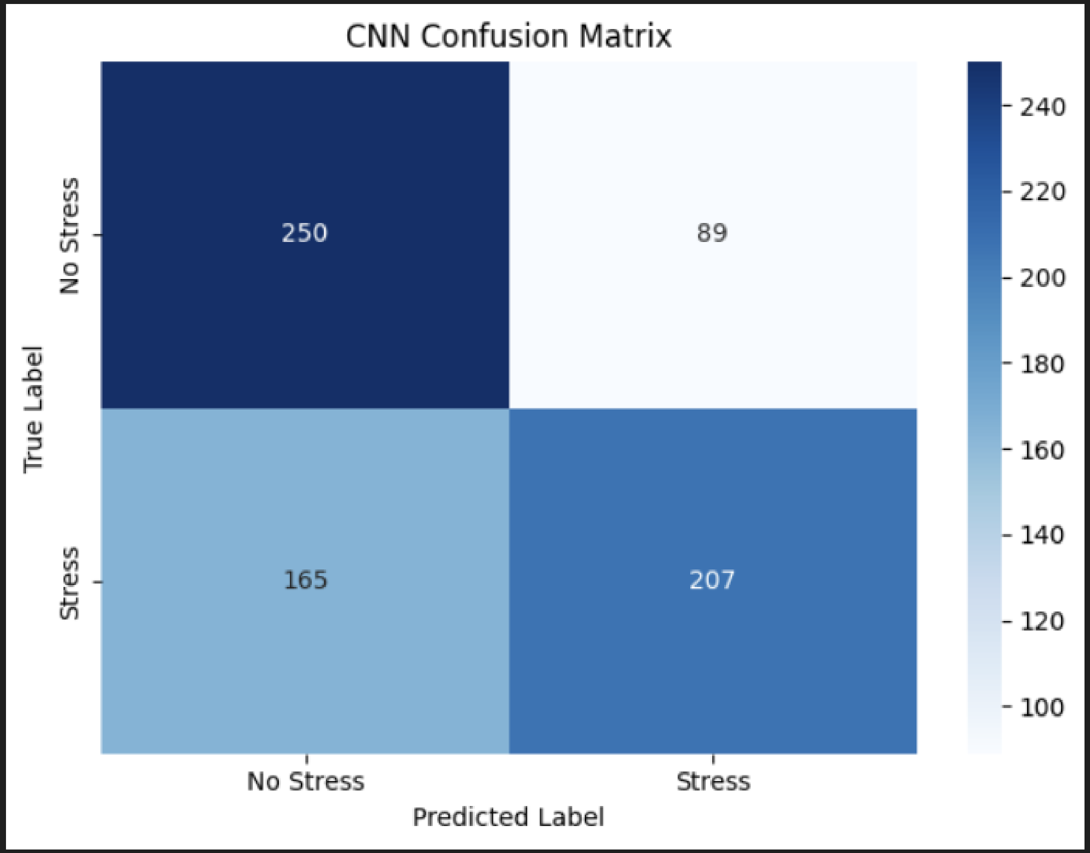
FINANCE SUBREDDITS

- r/assistance, r/homeless, r/almosthomeless, r/food_pantry
- Smaller, but most relevant to the research question

USAGE

- Full dataset used for training
- Finance subset reserved for targeted analysis

```
=== CNN Test Results ===
              precision    recall  f1-score   support

   No Stress       0.60      0.74      0.66       339
      Stress       0.70      0.56      0.62       372

    accuracy                           0.64       711
   macro avg       0.65      0.65      0.64       711
weighted avg       0.65      0.64      0.64       711
```

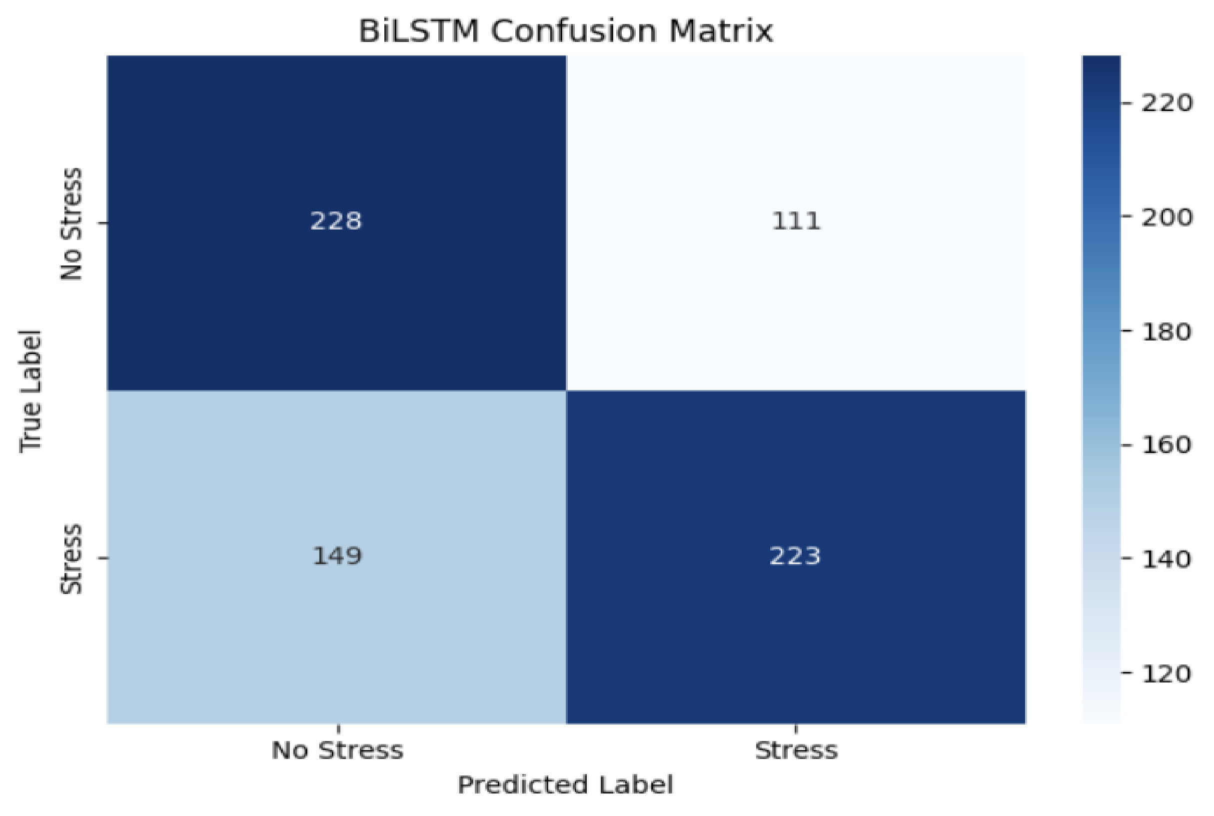


Methodology: CNN

- **Embedding layer:** Word2Vec weights, vocab 6,875 × 100 dims
- **Three parallel Conv1D filters:** kernels 2, 3, 4 → bigrams, trigrams, 4-grams
- **Max-over-time pooling:** strongest signal from each filter
- **Dropout (0.5)** → fully connected → 2 output classes

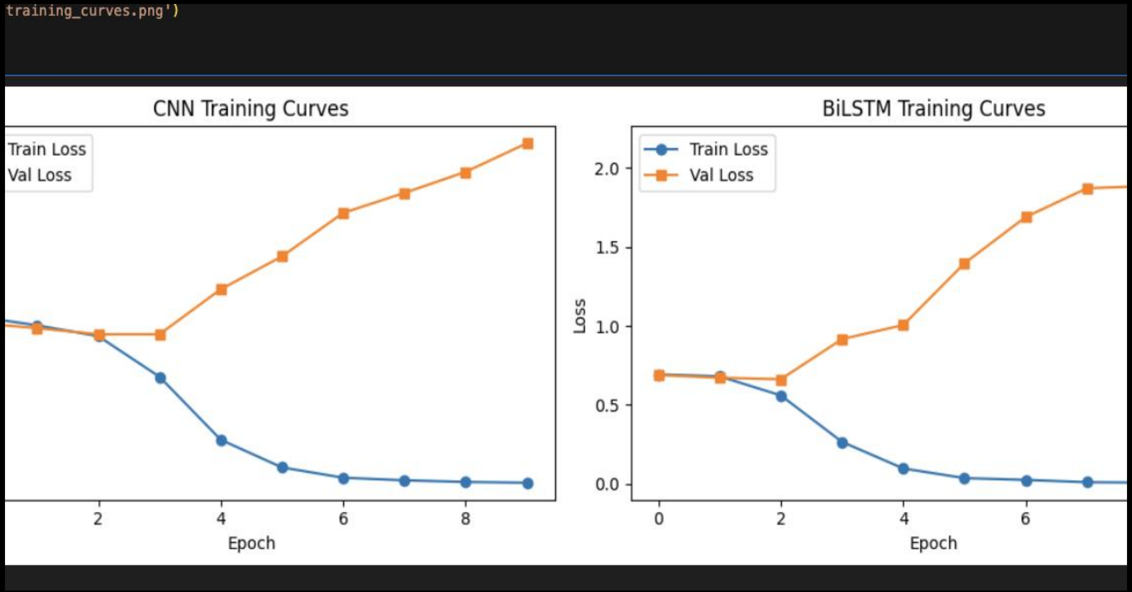
Methodology: BiLSTM

```
=== BiLSTM Test Results ===
              precision    recall  f1-score   support

   No Stress       0.60      0.67      0.64       339
      Stress       0.67      0.60      0.63       372

    accuracy                           0.63       711
   macro avg       0.64      0.64      0.63       711
weighted avg       0.64      0.63      0.63       711
```



- **Embedding layer:** same Word2Vec initialization as CNN
- **2-layer BiLSTM:** hidden dim 128, reads forward + backward
- **Long-range context:** better for narrative-style distress
- **Final hidden states** → concatenate → Dropout → output layer

Training Behavior: Overfitting in Both Models



	Model	Accuracy	F1 Score
0	CNN	0.6428	0.6404
1	BiLSTM	0.6343	0.6342

TRAINING DYNAMICS

Train loss → 0 by epoch 4–5

Validation loss diverges and rises — classic overfitting

PEAK VALIDATION ACCURACY

CNN: ~68% at epoch 7

BiLSTM: ~65% at epoch 9

TAKEAWAY

Both models memorize training data rather than generalize → motivates dataset and regularization limits

Model Performance & Financial Stress Subset

CNN

64.3% accuracy · F1 = 0.640

Best on all metrics

BILSTM

63.4% accuracy · F1 = 0.634

Trails CNN narrowly

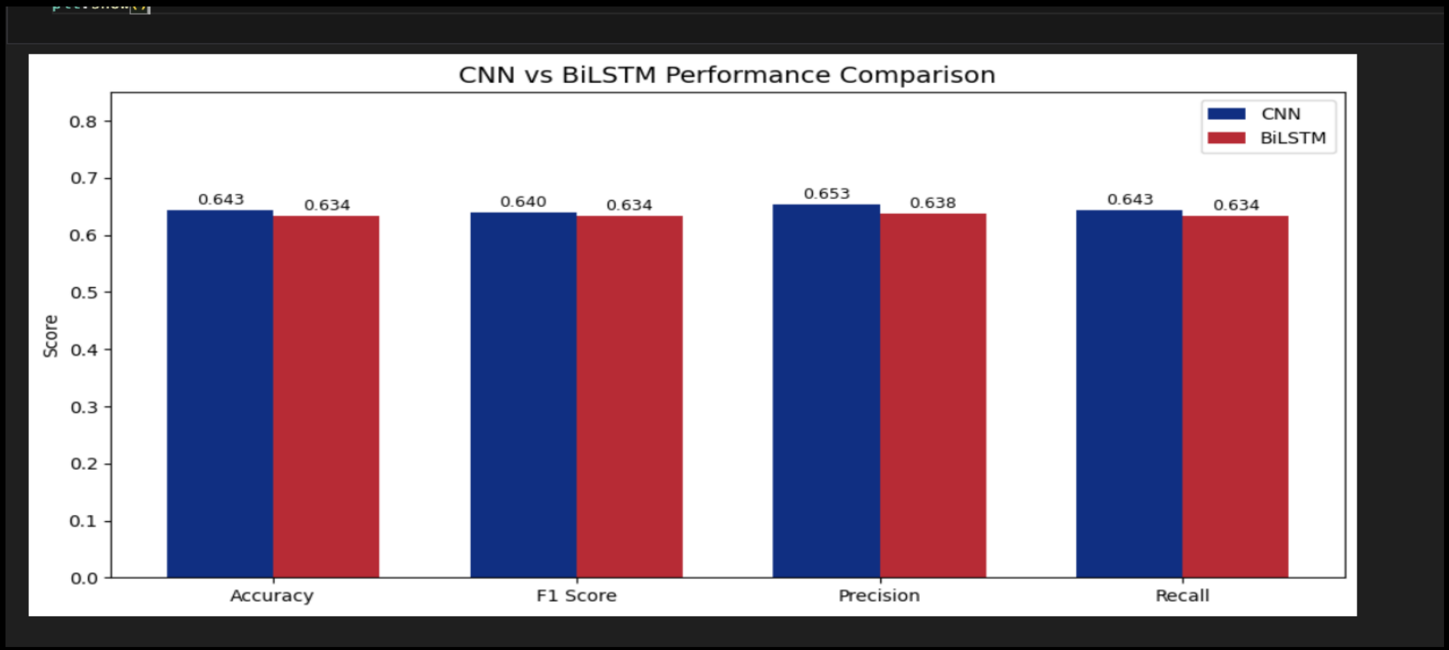
FINANCE SUBSET

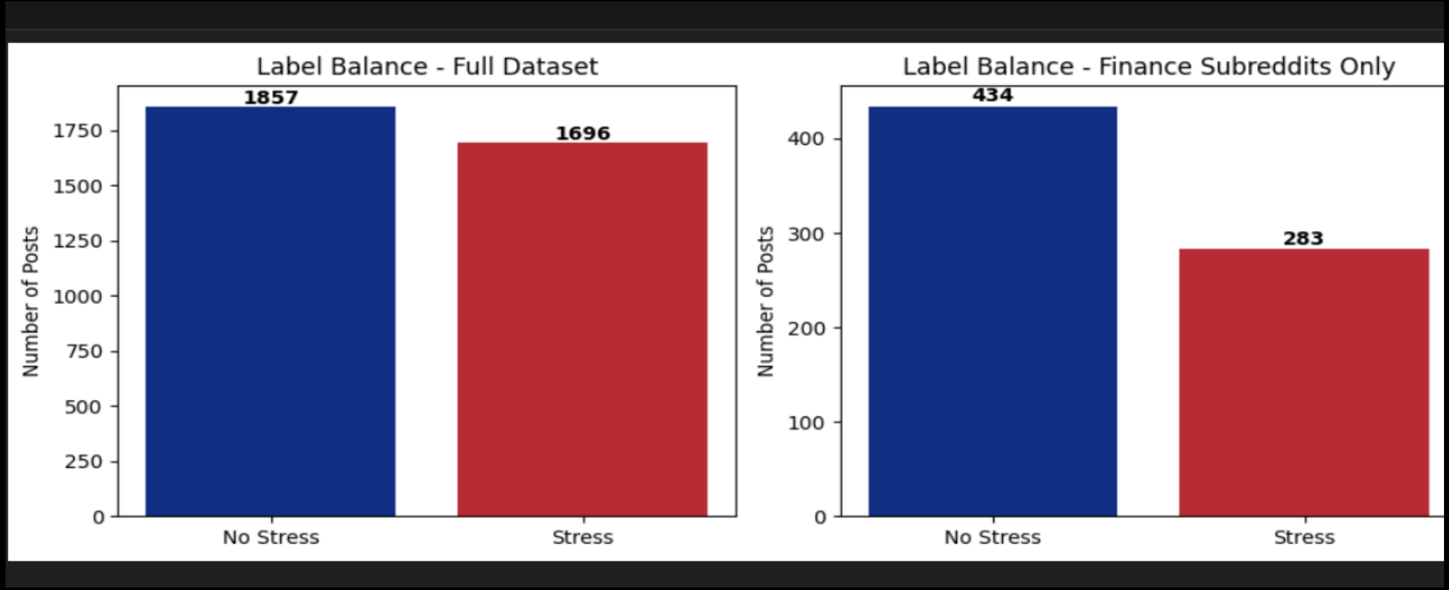
434 not-stressed vs. 283 stressed

Imbalance → harder than general stress

ERROR ANALYSIS

CNN: higher precision · BiLSTM: higher recall





Discussion & Error Analysis

- **False negatives:** indirect / subtle language slips past the model
- **False positives:** neutral mentions of anxiety flagged as stress
- **Root cause:** both models lean on keywords, not context
- **Domain skew:** finance subreddits have more “not stressed” posts → harder domain
- **CNN vs. BiLSTM:** BiLSTM trades precision for higher recall

Limitations

- **Small dataset:** ~2,500 training posts → overfitting and capped accuracy
- **Surface-level signals:** models detect keywords, not context or tone
- **Finance subset imbalance:** limits domain-specific learning
- **Dated corpus:** 2017–2018 data may not reflect current financial stress language

Conclusion

KEY FINDINGS

- **CNN edged out BiLSTM.** Local phrase patterns matter more than sequence for short Reddit posts.
- **Domain matters.** Financial stress is harder to detect than general stress.

FUTURE WORK

- **Cross-platform:** expand beyond Reddit to Twitter / X.
- **Pretrained models:** fine-tune BERT on finance-specific text.
- **Updated data:** refresh post-2022 to capture inflation-era language shifts.

The End! Thanks for watching!!

By Alasya Zeweldi